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GitHub Link	https://github.com/Yogeswar2906/MLA0405---DEEP-LEARNING/tree/main/Assessment/CO3

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING
SLOT : D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

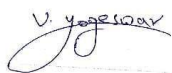
CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I V Yogeswar / 192425175 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

V Yogeswar
Name of the student:

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Total						

Signature of the Faculty:

Total Marks :

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
 2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
 3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
 4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
 5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
 6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.
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Case Study Question 1: CNN for Automated Medical Image Classification

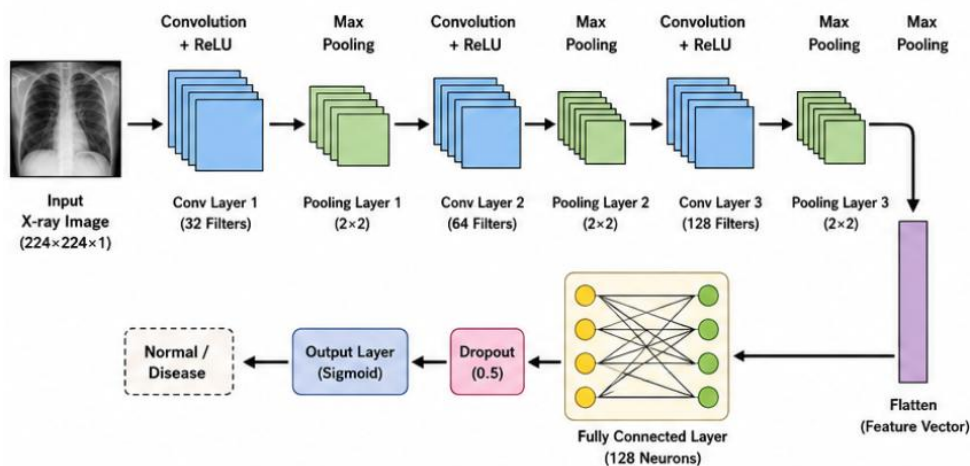
1. Suitable CNN Architecture

A suitable CNN architecture for chest X-ray classification can be:

Input → Convolution + ReLU → Max Pooling → Convolution + ReLU → Max Pooling → Convolution + ReLU → Max Pooling → Flatten → Fully Connected → Dropout → Output

- Input: Resized chest X-ray image, e.g., $224 \times 224 \times 1$.
- Convolutional layers: Extract features such as edges, textures and disease-related patterns.
- ReLU: Introduces non-linearity.
- Max Pooling: Reduces spatial dimensions while retaining important features.
- Flatten: Converts feature maps into a vector.
- Fully Connected layer: Performs final classification.
- Output layer: Uses sigmoid for two classes: Normal and Disease.

1. CNN Architecture for Medical Image Classification



2. Motivation and Role of CNN Components

CNNs are suitable because they automatically learn spatial features from images without manual feature extraction.

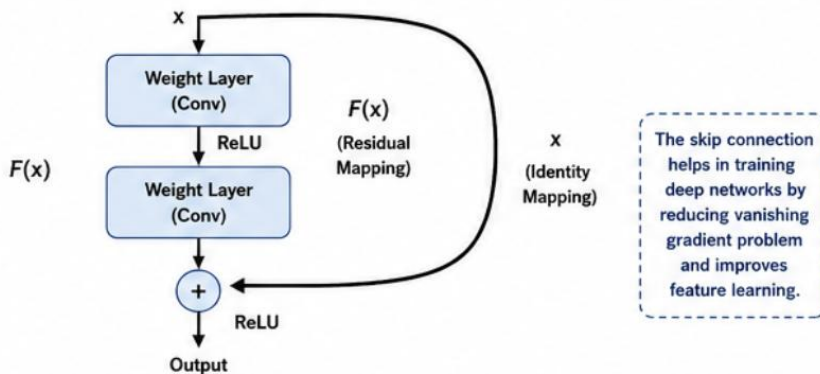
- Convolutional layers: Apply filters to detect features such as edges, textures and abnormal regions.
- Filters: Learn useful patterns from the X-ray images.
- Pooling layers: Reduce image dimensions and computational cost while retaining important information.
- Fully connected layers: Combine extracted features and perform classification.

3. Parameter Sharing

In a fully connected network, every neuron is connected to every input pixel, resulting in a very large number of parameters. In CNNs, the same filter weights are reused across different locations of the image. This is called parameter sharing.

For example, a 3×3 filter has only 9 weights (plus bias) and can scan the entire image. Therefore, CNNs require far fewer trainable parameters and have lower computational and memory requirements.

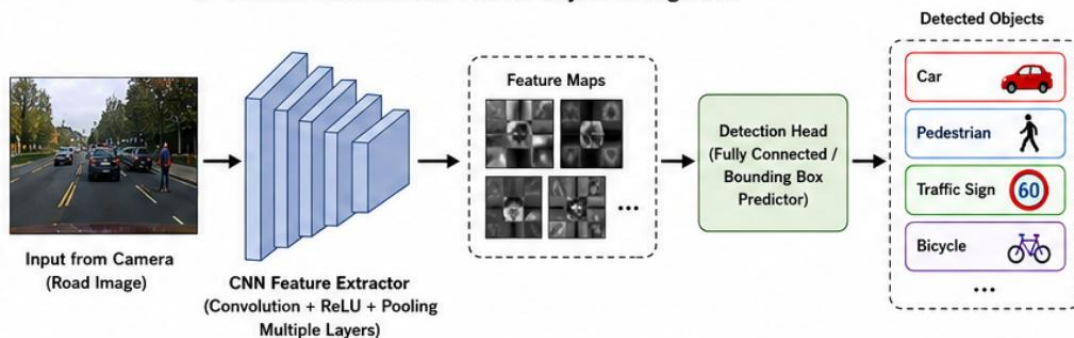
3. Skip / Residual Connection in ResNet



4. Regularization Techniques

- Dropout: Randomly disables some neurons during training, reducing overfitting.
- Data Augmentation: Creates additional training samples by applying transformations such as rotation, scaling, cropping and brightness changes.
- Batch Normalization: Normalizes intermediate activations, improving training stability and often providing a regularization effect.
- Early Stopping: Stops training when validation performance stops improving.

4. CNN for Autonomous Vehicle Object Recognition



5. AlexNet vs ResNet

ResNet is more appropriate for medical image classification.

- AlexNet: An older and relatively shallow CNN with 8 major layers.
- ResNet: Much deeper and uses residual/skip connections, which help solve vanishing-gradient and degradation problems.
- ResNet can learn more complex and discriminative features from medical images.
- ResNet generally provides better classification performance than AlexNet.
- Although deeper ResNet models require more computation, smaller versions such as ResNet-18 or ResNet-34 provide a good balance between accuracy and computational cost.

Therefore, ResNet is preferred for reliable chest X-ray disease classification.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition

6. CNN Architecture for Autonomous Vehicles

A suitable CNN architecture can be:

Input Image $\rightarrow$ Convolution $\rightarrow$ ReLU $\rightarrow$ Pooling $\rightarrow$ Convolution $\rightarrow$ ReLU $\rightarrow$ Pooling $\rightarrow$ Deeper Convolution Layers $\rightarrow$ Fully Connected/Detection Head $\rightarrow$ Output

The input consists of camera images from the vehicle. The CNN extracts visual features and identifies objects such as cars, pedestrians, bicycles, traffic signs and road obstacles.

For a practical autonomous-vehicle system, the CNN backbone can be combined with an object-detection head to determine both what the object is and where it is located.

7. Purpose of CNN Layers

- Input Layer: Receives camera images and performs preprocessing such as resizing and normalization.
- Convolution Layer: Applies filters to extract visual features.
- Activation (ReLU): Adds non-linearity and helps the network learn complex patterns.
- Pooling Layer: Reduces spatial dimensions and computational cost.
- Fully Connected/Output Layer: Produces the final class or detection results.

Feature learning occurs progressively:

Early layers → Edges and colors → Middle layers → Textures and shapes → Deep layers → Objects such as cars, pedestrians and signs.

8. Importance of Parameter Sharing

CNN filters use the same weights at different image locations. This is called parameter sharing.

It greatly reduces the number of trainable parameters and computational requirements compared with a fully connected network. It also allows the same feature, such as an edge or wheel shape, to be detected at different positions in an image. Thus, CNNs provide useful translation-related feature detection.

9. Overfitting and Regularization

Overfitting occurs when the CNN performs well on training images but poorly on unseen road images.

Suitable techniques include:

- Data augmentation: Rotation, cropping, scaling, brightness and contrast changes.
- Dropout: Randomly removes neurons during training.
- Batch normalization: Stabilizes training and can reduce overfitting.
- L2 regularization: Penalizes excessively large weights.
- Early stopping: Stops training when validation performance starts decreasing.

10. AlexNet vs ResNet

Feature	AlexNet	ResNet
Architecture	Traditional CNN	CNN with residual/skip connections
Depth	8 major layers	Can have 18, 34, 50, 101+ layers
Training	Relatively simple	Easier to train deep networks due to skip connections
Vanishing gradient	More affected	Skip connections greatly reduce the problem
Feature learning	Good basic features	Learns deeper and more complex features
Performance	Lower compared with modern networks	Generally higher accuracy
Efficiency	Relatively simple	Depends on version; ResNet-18/34 can be efficient

11. Recommended Architecture

ResNet is the better choice for the autonomous vehicle system because it provides stronger feature learning and generally better recognition accuracy than AlexNet. Its residual connections allow deeper networks to be trained effectively and reduce vanishing-gradient problems.

For real-time applications, a lightweight model such as ResNet-18 or ResNet-34 can provide a good balance between accuracy, computational cost and speed. Therefore, a lightweight ResNet-based architecture is recommended for autonomous vehicle object recognition.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
